Out-of-the-Box Performance of popular SDRs for EMC pre-compliance Measurements

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***Abstract*—This paper investigates the useability of popular software-defined radios for radiated electromagnetic pre-compliance measurements. Because of their versatility and low costs, the focus is put on transverse electromagnetic cells as test site in combination with *CISPR 16-1-1* requirements. Based on these assumptions, the most relevant performance aspects are derived and verified by measurements. Furthermore, certain disadvantages caused by the respective frequency conversion principles are examined. With the presented results, further improvements, implemented by simple modifications, are discussed. Finally, pros and cons of each device are contrasted.**

Index Terms*—Pre-compliance, electromagnetic interference (EMI), electromagnetic compliance (EMC), software-defined radio (SDR), *CISPR

I. Introduction

Nowadays, electromangetic compliance (EMC) has become a prevalent topic during prototyping phase and certification process. Because of changing requirements and electronic design challenges, it is often unclear if a project will pass the required tests. Hence, pre-compliance measurements are of high interest to minimize the risk for failing an examination. The aim of this paper is to analyze the capability of certain software-defined radios (SDRs) for estimating test results in advance. Due to space and shielding limitations, radiated emission measurements using an antenna are often not sufficient. Thus, transverse electromagetic (TEM) cells are a good supplement to circumvent these problems as they are typically compact in size and are not as prone to interference. In addition to this, the financial investment is low and it offers the ability to perform immunity tests as well [1]. It will be investigated how TEM cells affect the requirements for SDRs.

With the introduction of *CISPR 16-1-1*, a norm has been developed to identify the suitability of a receiver for emission measurements [2]. Based on the chosen environement, the focus will be put on radiated emission tests in *CISPR* band C and D, covering a frequency range from $30\,\mathrm{MHz}$ to $1\,\mathrm{GHz}$. The requirements for fully-compliant broadband time-domain receivers have been investigated in [3]. In order to further improve the understanding regarding the regulations for receivers with different bandwidths at the frontend and the analog-to-digital converter (ADC), the main mandatory aspects are analyzed and verified by measurements.

In previous works, a lot of effort has been put into the evaluation of pre-compliance equipment. Different low-cost tools to identify electromagnetic interference (EMI) with antennas are presented in [4]. Furthermore, in [5] and [6], a multi-probe EMC system has been examined, which can be utilized for antenna characterization or to estimate current distributions. Up to now, the use of SDRs for radiated EMC measurements with TEM cells has not been considered.

The chosen SDRs, listed in Table I, are of different budget, hardware architecture and signal processing capabilities. The frontend of the *LimeSDR* incorporates a purely direct conversion principle with tunable anti-aliasing filters (AAFs) in front of the ADC and three variable gain amplifiers. In terms of down-conversion, filtering, and amplification the *HackRF One* has a similar concept. The main difference is the additional up-conversion stage making it to a heterodyne radio. In contrast, the *BB60C* and the *TwinRX* use a super-heterodyne principle and the gain is adjusted by attenuators. In addition to this, the *TwinRX* has switch-able filters in front of the first mixer making it less vulnerable for saturation by broadband signals.

II. Performance Aspects

In the following, the main performance aspects of SDRs regarding *CISPR 16-1-1* are derived analytically. Firstly, the return loss (RL) and related uncertainties are analyzed. Secondly, the required receiver sensitivity for TEM cells is determined. Lastly, the dynamic range (DR) for continuous wave (CW) and broadband signals depending on certain ADC properties is contrasted to frontend capabilities. The utilized *CISPR* compliant intermediate frequency (IF) filter is modeled with a Gaussian shape having a $6\,\mathrm{dB}$ bandwidth of $120\,\mathrm{kHz}$, an impulse bandwidth $\mathrm{B_I}$ of $127.7\,\mathrm{kHz}$, and a noise bandwidth $\mathrm{B_N}$ of $90.3\,\mathrm{kHz}$.

A. Return Loss

It is assumed that the receiver under test is calibrated by an ideally matched signal source. The resulting measurement uncertainty ϵ, for the reflection coefficients of the SDR Γ_{SDR} and the TEM cell Γ_{TEM}, can be calculated by [7]:

$$\epsilon = 20\log_{10}\frac{1}{1-|\Gamma_{\mathrm{TEM}}||\Gamma_{\mathrm{SDR}}|}. \tag{1}$$

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TABLE I
INVESTIGATED SOFTWARE-DEFINED RADIOS

Parameters	***Great Scott Gadgets HackRF One***	***Lime microsystems LimeSDR***	***Signal Hound BB60C***	***Ettus Research USRP X310 TwinRX***
f_{RF}	1 MHz–6 GHz	100 kHz–3.8 GHz	9 kHz–6 GHz	10 MHz–6 GHz
$f_{\mathrm{s_{max}}}$	20 MSa/s	61.44 MSa/s	40 MSa/s	100 MSa/s
ADC res. / ENOB	8 Bit / 7 Bit	12 Bit / 9 Bit	14 Bit / 12 Bit	14 Bit / 11 Bit
Interface	USB 2.0	USB 3.0	USB 3.0	1G/10G Eth.
Budget	∼300€	∼300€	∼3,000€	∼11,000€

In Figure 1, the calculated error for different RLs is depicted. Obviously, the matching of the utilized equipment may cause significant deviations. In *CISPR 16-1-1*, the receiver RL is defined to be equal or larger than 10 dB without external attenuators applied. Thus, a maximum error smaller than 1 dB may occur if the TEM cell has a RL above 10 dB.

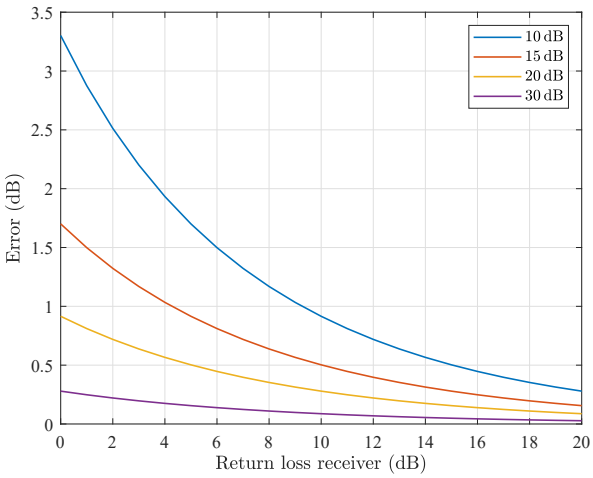


Fig. 1. Measurement uncertainty for different RLs of the receiver compared to TEM cell: 10 dB–30 dB.

B. Noise Floor

As this paper adresses indoor scenarios limited in space, the overall size of the TEM cell is a critical parameter. The *Teseq 500* has a septum height of 500 mm. Due to its length of approx. 3 m, this structure has been identified as maximum.

With the open-area test site (OATS) correlation algorithm for TEM cells it is possible to calculate the required sensitivity of the receiver [8]. Rearranging (2) allows to obtain the maximum received power $\sum V_i^2/Z_0$. The field factor e_{0_y} depends on the cell geometry and equipment under test (EUT) positioning. Moreover, $g_{\max}$ relates the OATS configuration to the cell and k_0 denotes the wavenumber. In Figure 2, a typical geometry for gigahertz transverse electromagnetic (GTEM) cells is visualized. The depcited parameters in conjuction with an OATS distance s of 10 m can be directly used to derive e_{0_y} and $g_{\max}$. In the following, it is assumed that one EUT position is dominant.

$$E_{\max} = g_{\max} \frac{60 k_0}{e_{0_y}} \sqrt{\frac{\sum_{i=1}^{3} V_i^2}{Z_0}} \qquad (2)$$

Applying (2) on a typical EMC standard, e.g., *CISPR 32*, the power level V^2/Z_0 is calculated [9]. In order to fulfill an uncertainty budget of 1 dB, a signal-to-noise ratio (SNR) of 6 dB is necessary. Furthermore, three scalar measurements are superposedd causing a noise enhancement of 5 dB. In total, a margin of 11 dB has to be subtracted for the receiver sensitivity, depicted in Figure 3. At 1 GHz, the noise floor is lowest with a level of −87 dBm. For a smaller cell size, the power level increases.

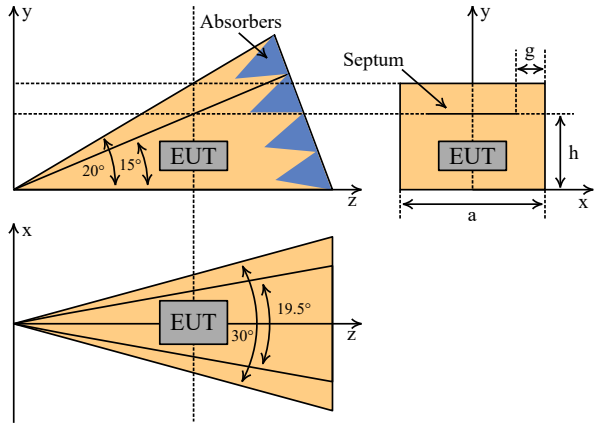


Fig. 2. Typical GTEM cell geometry, EUT centered at $(x, y) = (0, h/2)$.

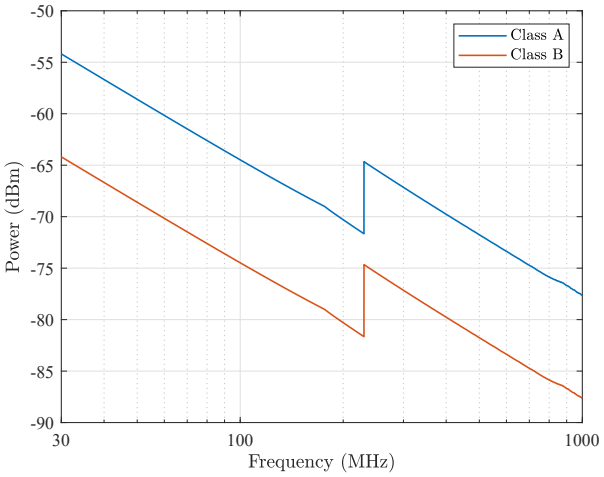


Fig. 3. Minimum receiver sensitivity for *CISPR 32*, $h = 433$ mm, $(x, y) = (0, h/2)$, $e_{0_y} = 16\sqrt{\Omega}$/m, OATS distance $s = 10$ m.

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C. Dynamic Range

With respect to *CISPR 16-1-1*, the DR for CW signals has to be at least 40 dB over the whole frequency range. Depending on the effective number of bits (ENOB), the baseband sample rate f_s and the noise bandwidth of the IF filter:

$$\mathrm{DR_{IF}} = \mathrm{ENOB} \cdot 6.02\,\mathrm{dB} + 1.76\,\mathrm{dB} + 10\log_{10}\left(\frac{f_s}{\mathrm{B_{N_{IF}}}}\right), \quad (3)$$

the achieved DR can be calculated. The quasi-peak detector increases the noise floor level by 5 dB and the peak-detector up to 13 dB [10]. In Figure 4, the ENOB required for CW signals is illustrated. With a minimum sampling-rate of 20 MSa/s the chosen SDRs, presumably, have no limitations in this case.

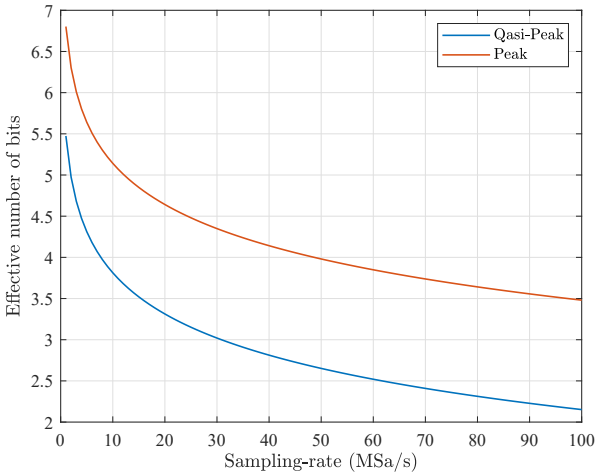


Fig. 4. Required ENOB for CW signals, utilizing different detectors, with respect to *CISPR 16-1-1*.

For broadband impulses, the required DR can be derived by including the impulse bandwidth. The maximum power of an impulse, with a strength of $v\tau$, results from $P_{\max} = (\sqrt{2}v\tau \mathrm{B_I})^2/Z_0$. The least significant bit voltage is $\Delta = \sqrt{2P_{\max}Z_0}/2^{b-1}$. Hence, the DR in IF and baseband domain can be related by the quantization noise power $\mathrm{P_{N_{ADC}}} = \Delta^2/(12Z_0)$. Consequently, with (4) the ADC requirements for broadband impulses are determined.

$$\mathrm{DR_{IF}} = 10\log_{10}\left(\frac{3}{2}\frac{f_s}{\mathrm{B_{N_{IF}}}}\left(2^{\mathrm{ENOB}}\frac{\mathrm{B_{I_{IF}}}}{\mathrm{B_{I_{ADC}}}}\right)^2\right) \quad (4)$$

The DR must be at least 19 dB in peak detection and 50 dB in quasi-peak detection mode to be capable of measuring isolated transients. In Figure 5, (4) is evaluated for the SDRs mentioned in Table I. It is shown, that the impulse bandwidth of the AAF can be reduced to compensate for a lower ENOB. Receivers with variable AAFs benefit from this behavior.

Taking the quasi-peak detector and the sensitivity at 1 GHz, derived in Section II-B, into account, the required power handling of the frontend can be calculated:

$$P = -87\,\mathrm{dBm} - 5\,\mathrm{dB} + \mathrm{DR_{IF}} + 20\log_{10}\frac{\mathrm{B_{I_{pre}}}}{\mathrm{B_{I_{IF}}}}. \quad (5)$$

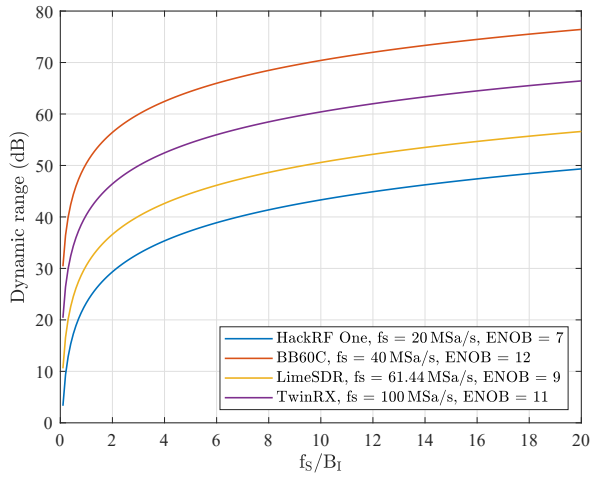


Fig. 5. Achievable DR for broadband impulses over the ratio of sampling-rate to impulse bandwidth of the AAF.

In Figure 6, the corresponding results, declared as 1 dB compression level, are depicted for a preselection impulse bandwidth equal to band D. It can be concluded that the DR of the frontend has to be increased if the AAF bandwidth is reduced and the signal level at the ADC is kept constant.

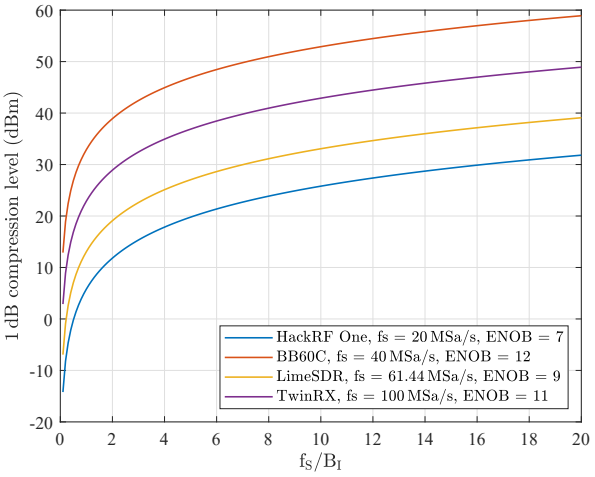


Fig. 6. 1 dB compression level of the frontend over the ratio of sampling-rate to impulse bandwidth of the AAF, $\mathrm{B_{I_{pre}}} = 700$ MHz.

III. Measurements

Based on Section II, the measurement results and performance degrading aspects, using the introduced IF filter, are presented. For PHY-layer configuration and data reception of the SDRs *Python* modules, i.e., *SoapySDR* have been utilized. Due to its simplicity, the graphical user-interface provided by *Signal Hound* was used for most of the measurements with the *BB60C*. Regarding the characterization of broadband impulses, the *Matlab* interface was considered. The *HackRF One* and the *LimeSDR* do not come with a surrounding shielding. As

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a lot of over-the-air interference occurs in *CISPR* bands C and D, they have been put into a metal-case with absorber sheets. Furthermore, all SDRs have been provided with a preselection filter[1] covering a bandwidth of 20 MHz–1 GHz. The configuration parameters of the SDRs can be found in Table II. The given AAF bandwidths of the SDRs have been verified of achieving an alias suppression larger than 40 dBc to be *CISPR* compliant.

TABLE II
SOFTWARE-DEFINED RADIO PARAMETERS

Para-meters	Software-defined radios			
	HackRF One	***LimeSDR***	***BB60C***	***TwinRX***
f_s	18 MSa/s	54 MSa/s	40 MSa/s	100 MSa/s
AAF	9 MHz	38 MHz	27 MHz	80 MHz
f_{IF}	−2.25 MHz	−2.25 MHz	0 Hz	20 MHz

It should be mentioned in advance that some measurement results are influenced by the non-ideal behavior of the SDRs. Especially, the characterization of DRs is sensitive to spurious emissions hidden in the noise floor or to the non-linear behavior of the implemented components. Consequently, discontinuities have to be expected.

A. Return Loss

The measured RL for certain gain configurations is illustrated in Figure 7. The gain has been adjusted to achieve a peak-detected noise level of −87 dBm at 1 GHz. Except the *LimeSDR*, all devices comply with *CISPR 16-1-1* having a RL larger than 10 dB. Referring to Figure 1, a maximum error larger than 3 dB has to be expected for this SDR.

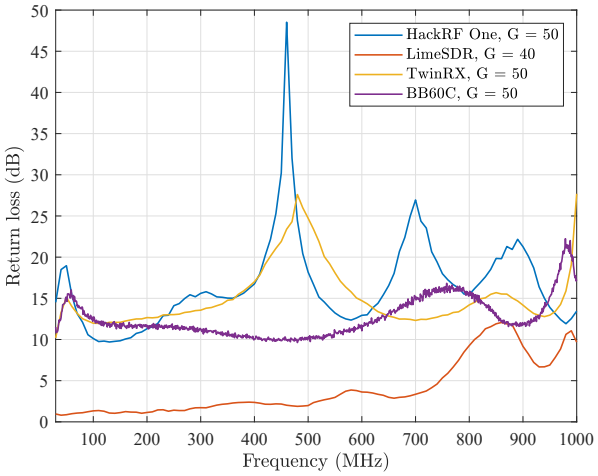


Fig. 7. Return loss over frequency for a certain gain configuration.

B. Noise Floor

The noise floor over *CISPR* bands C and D has been analyzed in peak-detection mode for a recording length of 1 s.

[1]Lumped LC bandpass filter: *Mini-Circuits ZABP-510-S+*

Comparing Figure 8 with the receiver sensitivity illustrated in Figure 3, it shows that all devices are fulfilling that requirement. The large amount of spurious emission can be reduced by shifting the IF. It must be mentioned that the user-interface of the *BB60C* already performs this task resulting in a cleaner spectrum.

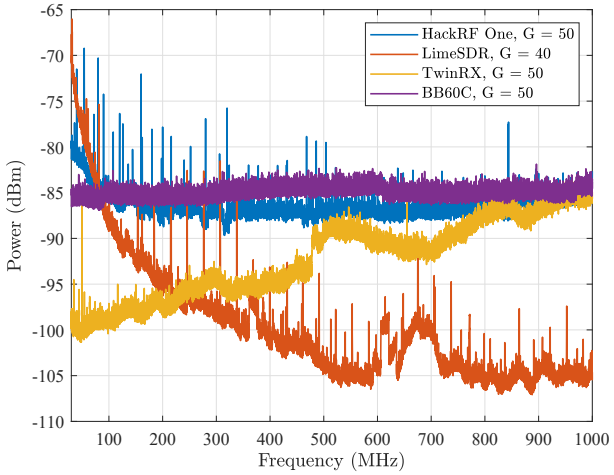


Fig. 8. Noise floor over frequency for a certain gain configuration in peak-detection mode with a recording length of 1 s.

Even if the noise floor of the SDRs seems to be a minor problem, it is not negligible in combination with the RL. Because of the reduced sensitivity, it is not possible to apply an attenuator for forced matching the *LimeSDR* in band C unless a smaller TEM cell is used.

C. Image Rejection

The *HackRF One* and the *LimeSDR* perform homodyne down-conversion and are operated as low IF receiver. In order to investigate gain dependencies of IQ-imbalances, the image rejection ratio (IRR) was measured. For low gain values, an SNR larger than 50 dB caused a saturation of the frontend. Hence, the target SNR was set to a range of 45–50 dB using the average detector. Comparing the results in Figure 9, the *LimeSDR* has a relatively small optimum gain range achieving an IRR larger than 40 dBc. Clearly, this SDR would benefit from an attenuator but as already stated, the sensitivity would be too low in band C for the investigated GTEM cell.

D. Harmonic Distortion

Due to harmonics of local oscillators (LOs), undesired frequencies are down-converted and interfere with the signal of interest. The *HackRF One*, *BB60C*, and *TwinRX* circumvent this problem by an up-conversion stage filtering out the harmonic frequency content. Hence, the *LimeSDR* is of special interest for this scenario. The respective results, with a leveled SNR between 45–50 dB, are illustrated in Figure 10. The odd-order LO products cause the most disturbance. Surprisingly, the 5th and 9th harmonic are even stronger than the fundamental one. For higher frequencies, the applied bandpass filter enhances the performance.

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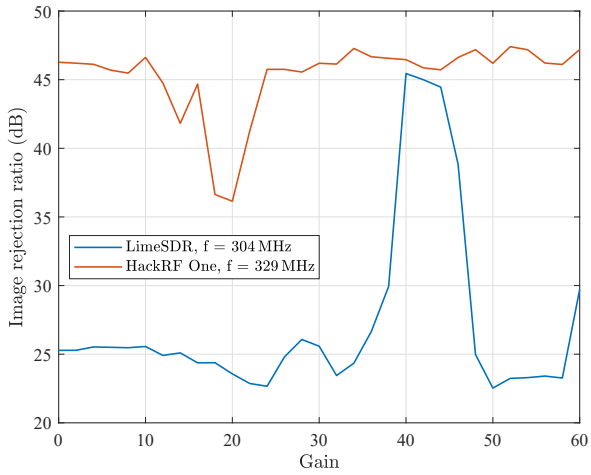


Fig. 9. IRR over gain using average detector for SDRs with a homodyne down-conversion stage.

Even though the performance of the *LimeSDR* is suffering significantly in *CISPR* band C, it is possible to gain information about the radiated emission of an EUT. By the analysis of band D, frequency content can be identified and avoided in band C by shifting the IF properly. If the distortion cannot be maintained in an acceptable range, additional preselection filtering is inevitable.

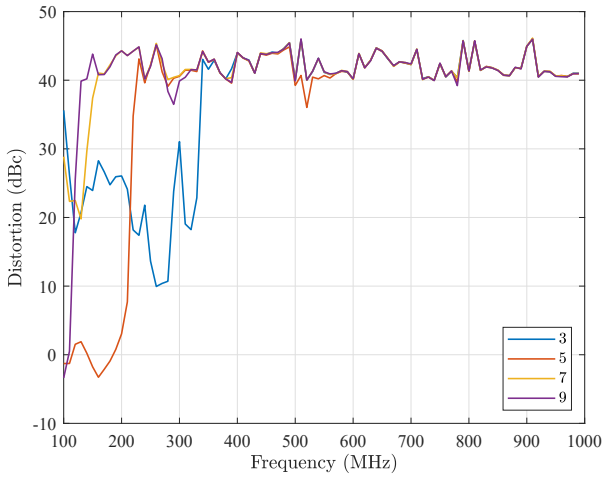


Fig. 10. Distortion due to LO feedthrough over frequency of the *LimeSDR* for odd-order harmonics $n = 3$–9 ($nf_{\mathrm{LO}} - f_{\mathrm{IF}}$).

E. Broadband Impulses

The DR for transients is considered by comparing the IF envelope of a CW signal with the maximum of an impulse in time-domain. For the impulse source (*Schwarzbeck IGU 2912*) calibration procedure, a fully-compliant EMI receiver (*Agilent MXE N9038A*) was utilized.

In the following, the measurement procedure to characterize the achievable DR is explained. For a given CW signal power, the SDR levels the desired SNR by altering the frontend gain and taking the average of the IF envelope. Then, impulses with a repetition rate of 100 Hz are measured. In order to get stable results, the local maxima of a sequence of transients is detected and averaged over a recording length of 10 s. The threshold for the detector is set five times higher than the standard deviation of the noise floor. As the impulse source is flat in frequency domain over *CISPR* bands C and D, the spectrum is limited with a low and highpass filter[2] for the respective range. Recapitulating the receiver sensitivity at 1 GHz, a CW power of $-87\,\mathrm{dBm} + 11\,\mathrm{dB} = -76\,\mathrm{dBm}$ is chosen for analyzing the achievable DR for impulses. The resulting error, depicted in Figure 11, is calculated by evaluating the ratio of the averaged impulse and CW samples. In Figure 12 and Figure 13, the

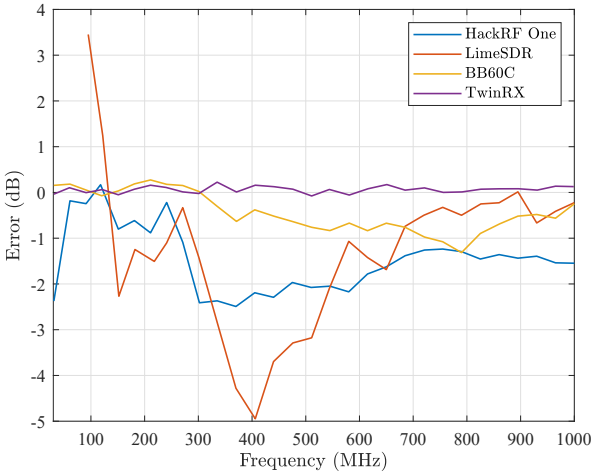


Fig. 11. Amplitude error between CW signal and impulses over frequency.

related DRs and adjusted gain values are illustrated. In order to guarantee an optimum performance in peak-detection mode a minimum DR of 20 dB is targeted. This goal implies an SNR of 6 dB and a crest-factor of 13 dB. Obviously, only the *BB60C* and the *TwinRX* stay in the error bound of ± 1.5 dB, required by *CISPR*. The remaining SDRs are only partially compliant. Furthermore, the *LimeSDR* cannot level the desired SNR for low frequencies anymore. Hence, the error at 30 MHz could not be validated. This behavior can be deduced from Figure 8. Maintaining a constant noise power level, the gain has to be increased for low frequencies. Besides the mentioned problems of LO harmonics and image rejection, the frontend may consequently get saturated causing the increasing error. For the *HackRF One*, the gain stays almost constant. Thus, the scenario has to be considered differently. The frontend up-converts the lower sideband and filters the spectrum with a lowpass filter. Therefore, the impulse bandwidth seen by the down-conversion stage increases for low frequencies. Consequently, this structure is potentially prone to an IF overload. Nevertheless, in case of a broadband impulse failing the EMC

[2] *Lowpass filter: Mini-Circuits ZX75LP-320-S+, highpass filter: Mini-Circuits ZX75HP-260-S+*

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test beyond 700 MHz, the *HackRF One* and the *LimeSDR* can measure the radiated power using the peak detector.

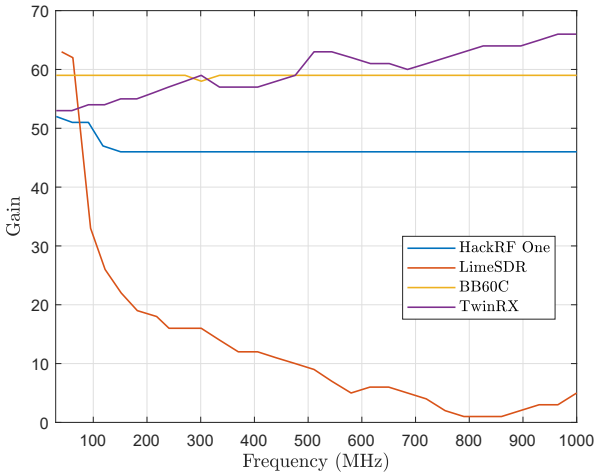


Fig. 12. Adjusted gain setting over frequency for impulse capability characterization.

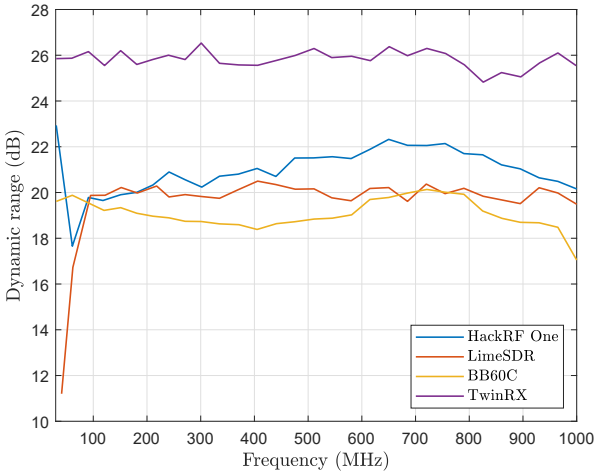


Fig. 13. Achieved DR over frequency for impulse capability characterization.

Estimating the performance using the quasi-peak detector, a margin for maintaining an error smaller than 1 dB and a noise enhancement factor each of 6 dB, have to be subtracted from the achieved DR. The *TwinRX* shows the best performance in weighting impulses down to a repetition rate of approx. 60 Hz, while the others are limited to about 200 Hz. Through the combination of the utilized high and bandpass filters, a power of approx. 0 dBm is received by the frontend of the SDR. Comparing the requirements from Section II-C, it shows that the performance is limited by the compression level of the SDR. Thus, the whole DR of the ADCs cannot be exploited.

IV. Conclusion

The suitability of SDRs for EMC pre-compliance measurements has been investigated in this paper. With an analytical approach the main performance aspects regarding TEM cells and *CISPR 16-1-1* were analyzed. The derived requirements have been verified by measurements and device dependent characteristics, degrading the performance, were demonstrated. The *LimeSDR* showed significant limitations in terms of RL, receiver sensitivity, IQ-imbalances, and LO feedthrough. Because of the reduced sensitivity in band C, the RL cannot be compensated by a forced match using attenuators for a GTEM cell with a septum height of 500 mm.

Regarding the measurement of broadband impulses using peak detection, the *TwinRX* and the *BB60C* showed the best performance. The maximum error bound of ± 1.5 dB could be met over the whole frequency range. For several reasons, the *HackRF One* and the *LimeSDR* could not meet this requirement for low frequencies of *CISPR* bands C and D. Besides the mentioned limitations of the *LimeSDR*, the *HackRF One* is prone to an IF overload causing this behavior. Consequently, they are only partially compliant. If measurements with the quasi-peak detector are performed, the repetition rate of occurring transients have to be known. Taking the achieved DR in peak-detection mode as reference, the *TwinRX* can weight periodic impulses down to 60 Hz and the remaining SDRs are limited to approx. 200 Hz.

In conclusion, SDRs can be utilized for EMC pre-compliance measurements with limitations. Comparing the investigated receivers, the *BB60C* shows a good trade-off between performance, software support and price. If periodic transients repeating slower than 200 Hz wants to be measured with the quasi-peak detector, the *TwinRX* has to be taken into account. Regarding the low-budget SDRs, the *HackRF One* out-performs the *LimeSDR* because of its heterodyne structure.

References

[1] F. Ferrero, A. López, M. Valledor, J. C. Campo, "An affordable EMC pre-compliance test lab for educationl purposes," IEEE Instrumentation & Measurement Magazine, Vol. 22, No. 5, pp. 57–63, October 2019.

[2] S. Braun, M. Aidam, P. Russer, "Development of a multiresolution time domain EMI measurement system that fulfills CISPR 16-1," International Symposium on Electromagnetic Compatibility, pp. 388–393, Vol. 2, August 2005.

[3] CISPR 16-1-1, "Specification for radio disturbance and immunity measuring apparatus and methods Part 1: Radio disturbance and immunity measuring apparatus - Measuring apparatus," 2015.

[4] Tektronix, "Low-cost EMI pre-compliance testing using a spectrum analyzer," [Online]. Available: https://download.tek.com/document/37A_60141_2_Low-cost_EMI_pre-compliance.pdf

[5] R. T. Sánchez, M. S. Castañer, L. J. Foged, "Use of software defined radio receivers for antenna measurements," 2019 Photonics & Electromagnetics Research Symposium - Spring, pp. 1862-1869, June 2019.

[6] R. T. Sánchez, M. S. Castañer, L. J. Foged, "EMC measurement system based on software defined radio and diagnostic techniques," 2019 Antenna Measurement Techniques Association Symposium, pp. 1-6, October 2019.

[7] Microwave mismatch error analysis, Hewlett-Packard Application note 56, 1967.

[8] IEC 61000-4-20, "Electromagnetic compatibility (EMC) – Part 4-20: Testing and measurement techniques – Emission and immunity testing in transverse electromagnetic (TEM) waveguides," 2010.

[9] CISPR 32, "Electromagnetic compatibility of multimedia equipment – Emission Requirements," 2015.

[10] D. B. Geselowitz, "Response of ideal radio noise meter to continuous sine wave, recurrent impulses, and random noise," in IRE Transactions on Radio Frequency Interference, vol. RFI-3, pp. 2–11, May 1961.

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